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Tsl12C

Tenkasi District

Common Half Yearly Examination - December 2025

Standard 12

CHEMISTRY

Time Allowed: 3.00 Hours Maximum Marks: 70

PART - I

I. Choose the best answer: **15×1=15**

1) The fraction of total volume occupied by the atoms in a simple cubic is

a) $\left(\frac{\pi}{4\sqrt{2}}\right)$ b) $\frac{\pi}{4}$ c) $\frac{\pi}{6}$ d) $\frac{\pi}{3\sqrt{2}}$

2) Time required for the reactant concentration to reach one half of its initial value is called

a) first order b) zero order c) half life period d) second order

3) The hydrogen ion concentration of a buffer solution consisting of a weak acid and its salts is given by

a) $[H^+] = Ka\frac{[acid]}{[salt]}$ b) $[H^+] = Ka[salt]$

c) $[H^+] = Ka[acid]$ d) $[H^+] = Ka\frac{[salt]}{[acid]}$

4) The number of electrons delivered at the cathode during electrolysis by a current of 1A in 60 seconds is (charge of electron = 1.6×10^{-19}C)

a) 6.22×10^{23} b) 6.022×10^{20} c) 3.75×10^{20} d) 7.48×10^{23}

5) **Match the following:**

A) V_2O_5 - i) High density poly ethylene

B) Ziegler - Natta - ii) PAN

C) Peroxide - iii) NH_3

D) Finely divided Fe - iv) H_2SO_4

	A	B	C	D		A	B	C	D
a)	(iv)	(i)	(ii)	(iii)	b)	(i)	(ii)	(iv)	(iii)
c)	(ii)	(iii)	(iv)	(i)	d)	(iii)	(iv)	(ii)	(i)

6) The metal which is used in packing material for food items

a) Zn b) Zr c) Al d) Au

7) The element that does not show contenation among the following p-block element is

a) Carbon b) Silicon c) Lead d) Germanium

8) The basicity of pyrophosphorus acid ($H_4P_2O_5$) is

a) 4 b) 2 c) 3 d) 5

9) The transition element which has only +3 oxidation state is

a) Ni b) Mn c) Cr d) SC

10) A complex in which the oxidation number of the metal is zero is

a) $K_4[Fe(CN)_6]$ b) $[Fe(CN)_3(NH_3)_3]$

c) $[Fe(CO)_5]$ d) Both (b) and (c)

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11) $OHCH_2 - CH_2OH$ on heating with periodic acid gives
a) Methanoic acid b) Glyoxal c) Methanal d) CO_2

12) The reagent used to distinguish between Acetaldehyde and Benzaldehyde is
a) Tollens reagent b) Fehling's solution
c) 2, 4 – dinitro phenyl hydrazine d) Semi carbazide

13) Anitine + benzoyl chloride $\xrightarrow{NaOH}$ $C_6H_5NHCOC_6H_5$ this reaction is known as
a) Friedel - Crafts reaction b) HVZ reaction
c) Schotten - Baumann reaction d) None of these

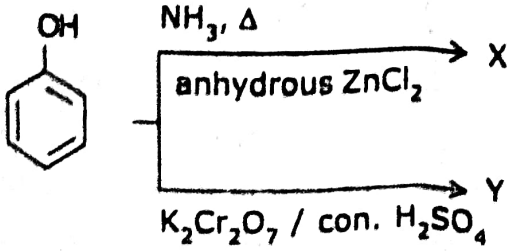
14) **Assertion** : A solution of sucrose in water is dextrorotatory. But on hydrolysis in the presence of little hydrochloric acid, it becomes levorotatory.
Reason : Sucrose hydrolysis gives equal amounts of glucose and fructose. As a result of this change in sign of rotation is observed.
a) If both assertion and reason are true and reason is the correct explanation of assertion.
b) If both assertion and reason are true but reason is not the correct explanation of assertion.
c) If assertion is true but reason is false.
d) If both assertion and reason are false.

15) The medicinal value of a drug is measured in terms of its
a) Deoxyribose b) Gold number
c) Therapeutic index d) Equilibrium constant

PART - II

Note: Answer any SIX questions. Question No. 24 is compulsory. **6×2=12**

16) Name the collector and depressing agent used in froth flotation process.
17) What type of hybridisation occur in (a) BrF_5 (b) BrF_3?
18) What are hydrate isomers? Explain with an example.
19) What are point defects?
20) State Ostwald's dilution law.
21) What are active centres?
22) How will you prepare Malachite green from benzaldehyde?
23) Write a short note on peptide bond.
24) Find the products X and Y in the following reactions.



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PART - III

Note: Answer any SIX questions. Question No. 33 is compulsory. **6×3=18**

25) Explain the principle of electrolytic refining with an example.

26) Give the uses of Borax.

27) Calculate the number of unpaired electrons in Ti^{3+}, Mn^{2+} and calculate the spin only magnetic moment.

28) Distinguish tetrahedral and octahedral voids.

29) Explain Pseudo first order reaction with an example.

30) Explain Kolbe's reaction.

31) $CH_3Br \xrightarrow{KCN} (A) \xrightarrow{H_3O+} (B) \xrightarrow{PCl_5} (C)$

Identify the product A, B and C.

32) How the tranquilizers work in body?

33) Calculate the molar conductance of 0.01M aqueous KCl solution at 25°C. The specific conductance of KCl at 25°C is 14.114×10^{-2} sm^{-1}.

PART - IV

Note: Answer ALL the questions. **5×5=25**

34) a) i) Describe a method for refining Titanium. **(3M)**

ii) Give the limitations of Ellingham diagram. **(2M)**

(OR)

b) i) How will you identify borate radical? **(3M)**

ii) CO is a reducing agent. Justify with an example. **(2M)**

35) a) Compare lanthanoids and actinoids. **(5M)**

(OR)

b) i) Give the difference between double salts and coordination compounds. **(3M)**

ii) What are interhalogen compounds? Give examples. **(2M)**

36) a) Calculate the percentage efficiency of packing in case of body centered cubic crystal. **(5M)**

(OR)

b) i) Derive the integrated rate law for a zero order reaction. **(3M)**

ii) Define Ionic product of water. **(2M)**

37) a) i) Describe adsorption theory of catalysis. **(3M)**

ii) Define gold number. **(2M)**

(OR)

b) How the following conversions are effected?

I) Phenol → Salicylaldehyde

II) Phenol → Phenolphthalein

III) Glycol → 1, 4 dioxane

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38) a) An aromatic aldehyde (A) of molecular formula C_7H_6O which has the smell of bitter almonds on treatment with $(CH_3CO)_2O$ and CH_3COONa to give compound (B) which is an aromatic unsaturated acid. (A) also reacts with (A) in the presence of at.KCN to give dimer (C). Identify (A), (B) and (C). Explain the reactions. **(5M)**

(OR)

b) i) How is chloropicrin prepared? **(2M)**

ii) What are the functions of lipids in living organism? **(3M)**

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